

UNIT II

INTRODUCTION OF HTML

Topics Covered

- 2.1 Structure of HTML
 - 2.2 Tags and Attributes of HTML
 - 2.3 Creating Web Pages Using Headings, Paragraph, BR and HR
 - 2.4 Image Tag
 - 2.5 Marquee Tag
-

2.1 STRUCTURE OF HTML

What is HTML?

HTML stands for **HyperText Markup Language**.

HTML is the standard markup language used to create and structure web pages. It tells a web browser how different elements of a web page should be organized and displayed.

HTML is **not a programming language**. It is a **markup language** because it uses markup tags to define the structure of content.

HTML can be used to create:

- Headings
- Paragraphs
- Images
- Hyperlinks
- Tables
- Lists
- Forms
- Audio
- Video
- Web page layouts

Basic Structure of an HTML Document

A basic HTML document has the following structure:

```
<!DOCTYPE html>
```

```
<html>
```

```
<head>
```

```
  <title>My First Web Page</title>
```

</head>

<body>

<h1>Welcome to My Web Page</h1>

<p>This is my first HTML web page.</p>

</body>

</html>

Explanation of HTML Structure

1. <!DOCTYPE html>

This declaration tells the browser that the document is an **HTML5 document**.

It is written at the beginning of the HTML document.

2. <html>

The <html> element is the **root element** of an HTML document.

All other HTML elements are generally placed inside the <html> element.

3. <head>

The <head> section contains information about the web page that is not normally displayed directly as page content.

It may contain:

- Page title
- Metadata
- CSS links
- JavaScript references
- Other information about the document

4. <title>

The <title> element defines the title of the web page.

The title is generally displayed in the browser tab.

Example:

<title>My First Web Page</title>

5. <body>

The <body> element contains the visible content of the web page.

It may contain:

- Headings
- Paragraphs
- Images
- Links
- Tables
- Forms
- Lists
- Videos

Example:

```
<body>

  <h1>Hello World</h1>

  <p>Welcome to HTML.</p>

</body>
```

Basic HTML Document Flow

The basic structure can be represented as:

HTML Document

↓

HTML Element

↓

HEAD + BODY

The **HEAD** contains information about the document.

The **BODY** contains the visible content displayed in the browser.

Example of a Complete HTML Page

```
<!DOCTYPE html>

<html>

  <head>

    <title>Student Information</title>

  </head>
```

```
<body>
```

```
<h1>Welcome to My Website</h1>
```

```
<p>
```

This website contains information about students.

```
</p>
```

```
</body>
```

```
</html>
```

Output

The browser displays:

Welcome to My Website

This website contains information about students.

2.2 TAGS AND ATTRIBUTES OF HTML

What are HTML Tags?

HTML tags are special keywords enclosed within **angle brackets** < >.

Tags are used to define and structure elements on a web page.

Example:

```
<h1>Welcome</h1>
```

Here:

- <h1> is the opening tag.
- Welcome is the content.
- </h1> is the closing tag.

The complete structure is called an **HTML element**.

General Structure

```
<tag>Content</tag>
```

Example:

<p>This is a paragraph.</p>

Types of HTML Tags

HTML tags can generally be classified into two types:

1. Container/Paired Tags
 2. Empty/Unpaired Tags
-

1. Container or Paired Tags

These tags have both an opening tag and a closing tag.

Example:

<p>This is a paragraph.</p>

Here:

- <p> is the opening tag.
- </p> is the closing tag.

Other examples:

<h1>Heading</h1>

Bold Text

<i>Italic Text</i>

<title>My Page</title>

2. Empty or Unpaired Tags

These tags do not require a closing tag because they do not contain content.

Examples:

<hr>

These are also called **void elements** in HTML.

What are HTML Attributes?

HTML attributes provide **additional information or properties** about an HTML element.

Attributes are written inside the **opening tag**.

General Syntax

```
<tag attribute="value">Content</tag>
```

Example:

```
<p align="center">Welcome</p>
```

Here:

- `<p>` is the tag.
- `align` is the attribute.
- `"center"` is the attribute value.

Another example:

```

```

Here:

- `src` specifies the image location.
- `alt` provides alternative text for the image.

Common HTML Attributes

1. id

The `id` attribute gives a unique identification to an HTML element.

Example:

```
<p id="intro">This is an introduction.</p>
```

2. class

The `class` attribute is used to assign one or more classes to an element.

It is commonly used with CSS and JavaScript.

Example:

```
<p class="important">Important Information</p>
```

3. style

The `style` attribute is used to apply inline CSS styling.

Example:

```
<p style="color: red;">This is red text.</p>
```

4. title

The title attribute provides additional information, often displayed as a tooltip when the user moves the mouse over the element.

Example:

```
<p title="This is additional information">
```

Move your mouse here.

```
</p>
```

5. href

The href attribute specifies the destination URL of a hyperlink.

Example:

```
<a href="https://www.google.com">Google</a>
```

6. src

The src attribute specifies the source or location of an external resource.

It is commonly used with images.

Example:

```

```

7. alt

The alt attribute provides alternative text for an image.

Example:

```

```

It is useful when the image cannot be displayed and also improves accessibility.

Difference Between HTML Tags and Attributes

HTML Tags	HTML Attributes
Define the structure of HTML elements. Provide additional information about elements.	
Written inside angle brackets.	Written inside the opening tag.
Example: <p>	Example: class="text"
Used to create elements.	Used to modify or describe elements.

2.3 CREATE WEB PAGE USING HEADINGS, PARAGRAPH, BR AND HR

HTML provides different tags to structure and organize the content of a web page.

The important tags in this section are:

- Heading tags
- Paragraph tag
- BR tag
- HR tag

A. Heading Tags

HTML provides six levels of headings.

They are:

`<h1>`

`<h2>`

`<h3>`

`<h4>`

`<h5>`

`<h6>`

`<h1>` is the largest and most important heading, while `<h6>` is the smallest heading.

Example

`<h1>This is Heading 1</h1>`

`<h2>This is Heading 2</h2>`

`<h3>This is Heading 3</h3>`

`<h4>This is Heading 4</h4>`

`<h5>This is Heading 5</h5>`

`<h6>This is Heading 6</h6>`

Heading Hierarchy

H1 → Main heading

H2 → Major subheading

H3 → Subsection

H4 → Smaller subsection

H5 → Smaller heading

H6 → Smallest heading

Example of Heading Web Page

```
<!DOCTYPE html>

<html>

<head>

  <title>Heading Example</title>

</head>

<body>

  <h1>Welcome to My Website</h1>

  <h2>About Us</h2>

  <h3>Our Services</h3>

  <h4>Web Development</h4>

  <h5>HTML Development</h5>

  <h6>Basic HTML</h6>

</body>

</html>
```

B. Paragraph Tag

The <p> tag is used to define a paragraph.

Syntax

```
<p>Content of the paragraph.</p>
```

Example

```
<p>
```

HTML is used to create and structure web pages.

```
</p>
```

Multiple paragraphs can be created using multiple <p> elements.

Example:

```
<p>This is the first paragraph.</p>
```

```
<p>This is the second paragraph.</p>
```

The browser normally places space between separate paragraphs.

C. BR Tag

The
 tag stands for **Line Break**.

It is used to move the following content to a new line.

The
 tag is an empty or void element.

Syntax

```
<br>
```

Example

```
<p>
```

My name is Umakant.

I am a student.

I am studying Computer Science.

```
</p>
```

Output

My name is Umakant.

I am a student.

I am studying Computer Science.

The
 tag is useful when a line break is required without starting a new paragraph.

D. HR Tag

The <hr> tag stands for **Horizontal Rule**.

It is used to create a thematic break or horizontal line between sections of content.

The <hr> tag is also a void element.

Syntax

```
<hr>
```

Example

```
<h1>About Us</h1>
```

```
<p>We provide web development services.</p>
```

```
<hr>
```

```
<h1>Contact Us</h1>
```

```
<p>Contact us for more information.</p>
```

The <hr> tag visually separates the two sections.

Complete Example Using H1, H2, P, BR and HR

```
<!DOCTYPE html>

<html>

<head>
  <title>My Web Page</title>
</head>
<body>
  <h1>Welcome to My Website</h1>
  <hr>
  <h2>About Me</h2>
  <p>
    My name is Umakant.<br>
    I am a Computer Science student.<br>
    I am interested in Web Development.
  </p>
  <hr>
  <h2>My Education</h2>
  <p>
    I am currently pursuing my degree in Computer Science.
  </p>
</body>
</html>
```

2.4 IMAGE TAG

What is the Image Tag?

The tag is used to display an image on a web page.

It is an **empty or void element**, so it does not require a closing tag.

Basic Syntax

```

```

Important Attributes of Image Tag

1. src

The src attribute specifies the location or path of the image.

Example:

```

```

If the image is stored inside a folder:

```

```

2. alt

The alt attribute provides alternative text for an image.

Example:

```

```

The alternative text is useful when:

- The image cannot be loaded.
 - The user uses a screen reader.
 - The image is unavailable.
-

3. width

The width attribute specifies the width of the image.

Example:

```

```

4. height

The height attribute specifies the height of the image.

Example:

```

```

Example Using Image Tag

```
<!DOCTYPE html>
```

```
<html>
```

```
<head>

  <title>Image Example</title>

</head>

<body>

  <h1>My Profile</h1>

  

</body>

</html>
```

Image Path

The image source can be specified using different paths.

1. Image in Same Folder

```

```

2. Image in a Subfolder

```

```

3. Image from Another Location

```

```

4. Image from a Web URL

```

```

Important Points About Image Tag

- `<img>` is used to display images.
- It is an empty/void element.
- The `src` attribute specifies the image source.

- The alt attribute provides alternative text.
 - width controls image width.
 - height controls image height.
 - Images can be stored locally or loaded from a URL.
-

2.5 MARQUEE TAG

What is Marquee Tag?

The <marquee> tag was used in older versions of HTML to create scrolling or moving text and images on a web page.

It allowed content to move:

- Left
- Right
- Up
- Down

Basic Syntax

```
<marquee>
```

```
    Welcome to My Website
```

```
</marquee>
```

The text will scroll across the web page.

Marquee Direction

The direction attribute specifies the direction in which the content moves.

1. Left

```
<marquee direction="left">
```

```
    Welcome
```

```
</marquee>
```

2. Right

```
<marquee direction="right">
```

```
    Welcome
```

```
</marquee>
```

3. Up

```
<marquee direction="up">
```

```
  Welcome
```

```
</marquee>
```

4. Down

```
<marquee direction="down">
```

```
  Welcome
```

```
</marquee>
```

Marquee Behavior

The behavior attribute controls how the marquee moves.

1. Scroll

The content continuously moves across the screen.

```
<marquee behavior="scroll">
```

```
  Welcome
```

```
</marquee>
```

2. Slide

The content moves and stops when it reaches the edge.

```
<marquee behavior="slide">
```

```
  Welcome
```

```
</marquee>
```

3. Alternate

The content moves back and forth.

```
<marquee behavior="alternate">
```

```
  Welcome
```

```
</marquee>
```

Marquee Scroll Amount

The scrollamount attribute controls the speed of movement.

Example:

```
<marquee scrollamount="10">
```

```
  Fast Moving Text
```

</marquee>

A higher value generally results in faster movement.

Marquee Scroll Delay

The scrolldelay attribute specifies the delay between movements.

Example:

```
<marquee scrolldelay="100">
```

Moving Text

```
</marquee>
```

Marquee Width and Height

The width attribute controls the width of the marquee area.

The height attribute controls its height.

Example:

```
<marquee width="500" height="100">
```

Welcome to My Website

```
</marquee>
```

Complete Marquee Example

```
<!DOCTYPE html>
```

```
<html>
```

```
<head>
```

```
  <title>Marquee Example</title>
```

```
</head>
```

```
<body>
```

```
  <h1>Welcome to My Website</h1>
```

```
  <marquee direction="left" behavior="scroll" scrollamount="10">
```

Welcome to our website!

```
  </marquee>
```

```
</body>
```

```
</html>
```

Important Note About Marquee

The <marquee> element is **obsolete and deprecated in modern HTML**. It is not recommended for use in modern web development.

Modern websites generally use:

- CSS animations
- JavaScript
- CSS transitions

instead of <marquee>.

However, the <marquee> tag may still be included in academic syllabi to teach older HTML features, so it is important to understand its syntax and attributes for examination purposes.

UNIT II – QUICK REVISION

- **HTML:** HyperText Markup Language used to create and structure web pages.
- **DOCTYPE:** Declares the HTML version/document type.
- **HTML:** Root element of an HTML document.
- **HEAD:** Contains metadata and information about the document.
- **TITLE:** Defines the browser tab title.
- **BODY:** Contains the visible content of the web page.
- **Tag:** Defines an HTML element.
- **Attribute:** Provides additional information about an HTML element.
- **Paired Tag:** Has an opening and closing tag.
- **Empty/Void Tag:** Does not require a closing tag.
- **H1–H6:** HTML heading tags.
- **P:** Used to create paragraphs.
- **BR:** Used to create a line break.
- **HR:** Creates a horizontal/thematic break.
- **IMG:** Used to display images.
- **SRC:** Specifies the image source.
- **ALT:** Provides alternative text for an image.
- **Marquee:** An obsolete HTML element previously used for scrolling content.
- **Direction:** Specifies the direction of marquee movement.

- **Behavior:** Specifies how marquee content moves.
 - **Scrollamount:** Controls marquee movement speed.
-

IMPORTANT EXAM QUESTIONS

1. What is HTML? Explain the basic structure of an HTML document.
2. Explain the purpose of `<!DOCTYPE html>`, `<html>`, `<head>`, `<title>`, and `<body>` tags.
3. Write and explain the basic structure of an HTML document.
4. What are HTML tags? Explain with examples.
5. What are HTML attributes? Explain with examples.
6. Differentiate between HTML tags and HTML attributes.
7. Explain paired and unpaired/void HTML tags with examples.
8. Explain the different heading tags in HTML.
9. Explain the `<p>` tag with an example.
10. What is the use of the `<br>` tag? Explain with an example.
11. What is the use of the `<hr>` tag? Explain with an example.
12. Write an HTML program to create a web page using headings, paragraphs, `<br>`, and `<hr>` tags.
13. What is the `<img>` tag? Explain its important attributes.
14. Explain the `src`, `alt`, `width`, and `height` attributes of the image tag.
15. Write an HTML program to display an image on a web page.
16. Explain different ways of specifying an image path in HTML.
17. What is the marquee tag? Explain its attributes.
18. Explain the `direction`, `behavior`, `scrollamount`, and `scrolldelay` attributes of the marquee tag.
19. Write an HTML program to display scrolling text using the marquee tag.
20. Write a complete HTML program that uses:
 - Headings
 - Paragraphs
 - BR tag
 - HR tag
 - Image tag
 - Marquee tag

21. Write short notes on:

- HTML Tags
- HTML Attributes
- Heading Tags
- Paragraph Tag
- BR Tag
- HR Tag
- Image Tag
- Marquee Tag